NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES

Marks: 100	**ASSIGNMENT-4**	**Dead Line: 28-03-26**

Subject: Probability and Statistics **Sections: ALL**
Instructor: Mr. Hazber Samson, Miss Javeria, Miss Warda Date: 23-03-2026

Q1. Suppose a television contains 20 transistors, 2 of which are defectives. Five transistors are selected at random, removed and inspected.
(a) probability of selecting no defectives,
(b) probability of selecting at least one defective.
(c) probability of selecting at most two defective.
(d) The mean and variance for the number of defectives selected.

Q2. (a) The probability to be caught while running a red light is estimated as 0.1. What is the probability that a person is first caught on his 10th attempt to run a red light?
(b) From past experience it is known that 3% of accounts in a large accounting population are in error. What is the probability that the first account in error is found on the 5th try?

Q3. A computing center is interviewing people until they find a qualified person to fill a vacant position. The probability that any single applicant is qualified is 0.15.
(a) Find the expected number of people to interview.
(b) Find the probability the center will need to interview between 4 and 8 people (both inclusive).

Q4. (a) Jim is a high school baseball player. He has 0.25 batting average, meaning that he makes a hit in 25% of his tries (\at-bats"). What is the probability that Jim makes his second hit of the season on his sixth at-bat? Also find expectation and variance.
(**b)** A telemarketer needs to sell 3 insurance policies before lunch. He estimates the probability of a sale as 0.1. How many calls, on average, does he need to make before lunch? What is the probability that he needs exactly 25 calls to reach his goal?

Q5. Number of flaws in a disk follows a Poisson distribution with the mean 0.5. If 5 disks are selected independently,
(a) what is the probability that none of the disks have flaws?
(b) what is the probability that all 5 have a total of 2 flaws?
(c) what is the probability that total of at least 2 flaws?

Q6. Let X be a discrete random variable following the Geometric Distribution having p.m.f

$$P(X = x) = p\,q^{x-1} \quad , x = 1,2,3,....$$

(a) Show that sum of probabilities equals to unity.
(b) Show that mean of Geometric Distribution is given by $1/p$
(c) Show that variance of Geometric Distribution is given by q/p^2
(d) Show that Moment Generating Function (MGF) is given by $pe^t/(1-qe^t)$
also find mean and variance using MGF.
(e) Show that Cumulant Generating Function (CGF) is given by $\ln(pe^t)-\ln(1-qe^t)$
also find mean and variance using CGF.

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Q7. Let X be a continuous random variable following the Beta Distribution having p.d.f

$$f(x)=\begin{cases}\dfrac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}x^{\alpha-1}(1-x)^{\beta-1} & for\ 0<x<1\\ 0 & otherwise\end{cases}$$

(a) Show that area under the Beta curve is unity.
(b) Find the mean of Beta distribution.
(c) Find the variance of Beta distribution
(d) Find the Moment Generating Function (MGF) of Beta Distribution.

Q8. Let X be a continuous random variable following the Normal Distribution having p.d.f

$$f(x)=\frac{1}{\sigma\sqrt{2\pi}}e^{\frac{-1}{2\sigma^2}(x-\mu)^2}\quad -\infty<x<\infty$$

(a) Show that area under the normal curve unity.
(b) Show that mean of Normal Distribution is given by μ.
(c) Show that variance of Normal Distribution is given by σ^2.
(d) Show that Moment Generating Function (MGF) is given by $M_X(t)=e^{\mu t+\frac{1}{2}t^2\sigma^2}$ also find mean and variance using MGF.
(e) Show that Cumulant Generating Function (CGF) is given by $K_X(t)=\mu t+\frac{1}{2}t^2\sigma^2$ also find mean and variance using CGF.

Q9. Starting at 5:00 A.M., every half hour there is a flight from San Francisco airport to Los Angeles International airport. Suppose that none of these planes is completely sold out and that they always have room for passengers. A person who wants to fly to L.A. arrives at the airport at a random time between 8:45 A.M. and 9:45 A.M. Find the probability that she waits
(a) at most 10 minutes
(b) at least 15 minutes

Q10. The heights of boys at particular age follows a normal distribution with mean 150.3 cm and variance 25 cm. Find the probability that a boy picked at random from this age group
a. less than 153 cm,
b. more than 158 cm
c. between 150 cm and 158 cm.